


Engineering Electromagnetics
工程电磁场

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9: Conductors & Dielectrics

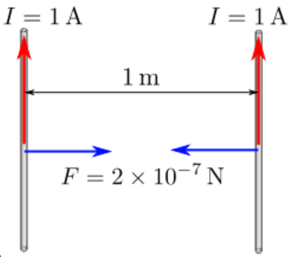
- ## Current and Current Density 电流和电流密度

Current is a flux quantity and is defined as: $I = \dfrac{dQ}{dt}$

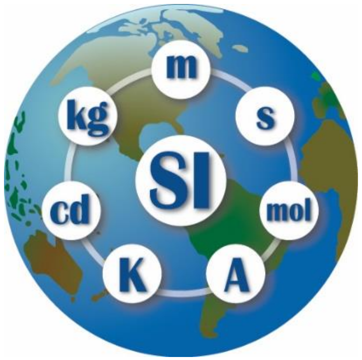
Current is defined as the motion of positive charges.
The unit of current is the ampere (A), and Current is a scalar.

Ampere is one of the seven SI unit symbol.
The constant current that will produce an attractive force of 2×10^{-7} newton per meter of length between two straight, parallel conductors of infinite length and negligible circular cross section placed one meter apart in a vacuum

André-Marie Ampère
(1775-1836)



国际计量局



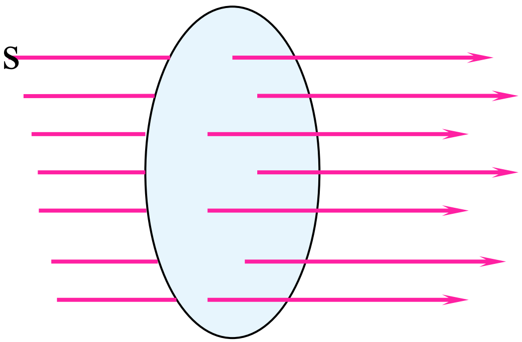
Meter Convention
米制公约

Metric Units (SI / MKS)

Quantity	SI	Abbr.	U.S. Customary	Abbr.
Base Units				
Mass	kilogram	kg	2.205 pounds	lbs
Length	meter	m	3.281 feet	ft
Time	second	s		
Electric Current	ampere	A		
Temperature	kelvin	K	1.8 Fahrenheit	ºF
Luminous Intensity	candela	cd		
Amount of Substance	mole	mol		
Derived Units				
Area	square meters	m^2	1.196 square yards	yd^2
Volume	cubic meters	m^3	35.32 cubic feet	ft^3
Density	kg per cubic meter	kg/m^3	0.06243 lbs / cubic foot	lbs/ft^3
Velocity	meters per second	m/s	2.237 miles per hour	mph
Acceleration	m/s per second	m/s^2	3.281 feet per sq. second	ft/s^2
Force	newton	$N = (kg{\cdot}m)/s^2$	0.2248 pound-force	lb_f
Energy or Work	joule	$J = (kg{\cdot}m^2)/s^2$	0.7376 foot-pounds	$ft{\cdot}lb_f$
Power	watt	$W = J{\cdot}s$	0.001341 horsepower	hp
Pressure	pascal	$Pa = N/m^2$	145.04×10^{-6} lbs. per sq. inch	psi

9: Conductors & Dielectrics

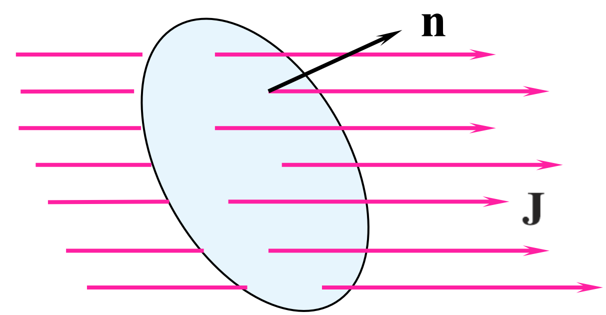
• Current and Current Density 电流和电流密度



ΔS

Current density, J, measured in A/m^2 , yields current in Amps when it is integrated over a cross-sectional area. The assumption would be that the direction of J is normal to the surface, and so we would write: $\Delta I = J_N \Delta S$

In reality, the direction of current flow may not be normal to the surface in question, so we treat current density as a vector, and write the incremental flux through the small surface in the usual way: $\Delta I = \mathbf{J} \cdot \Delta \mathbf{S}$ where $\Delta\mathbf{S} = \mathbf{n}\, da$



ΔS

Then, the current through a large surface is found through the flux integral:

$$\boxed{I = \int_S \mathbf{J} \cdot d\mathbf{S}}$$

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• Relation of Current to Charge Velocity

Consider a charge ΔQ, occupying volume Δv, moving in the positive x direction at velocity v_x

In terms of the volume charge density, we may write: $\Delta Q = \rho_v \Delta v = \rho_v \, \Delta S \, \Delta L$

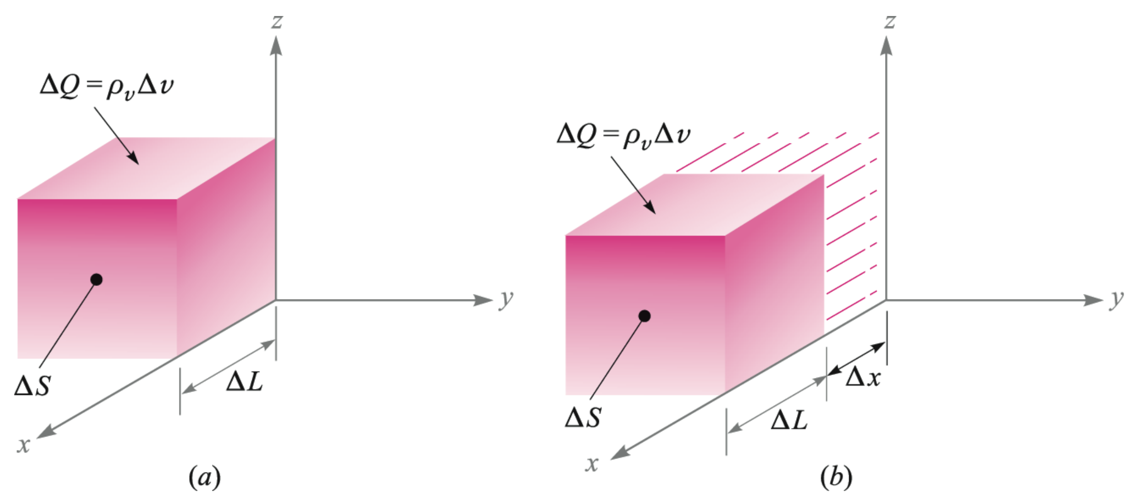
Suppose that in time Δt, the charge moves through a distance $\Delta x = \Delta L = v_x \, \Delta t$

Then $\Delta Q = \rho_v \, \Delta S \, \Delta x$

The motion of the charge represents a current given by:

$$\Delta I = \frac{\Delta Q}{\Delta t} = \rho_v \, \Delta S \frac{\Delta x}{\Delta t}$$

or $\Delta I = \rho_v \, \Delta S \, v_x$



9: Conductors & Dielectrics

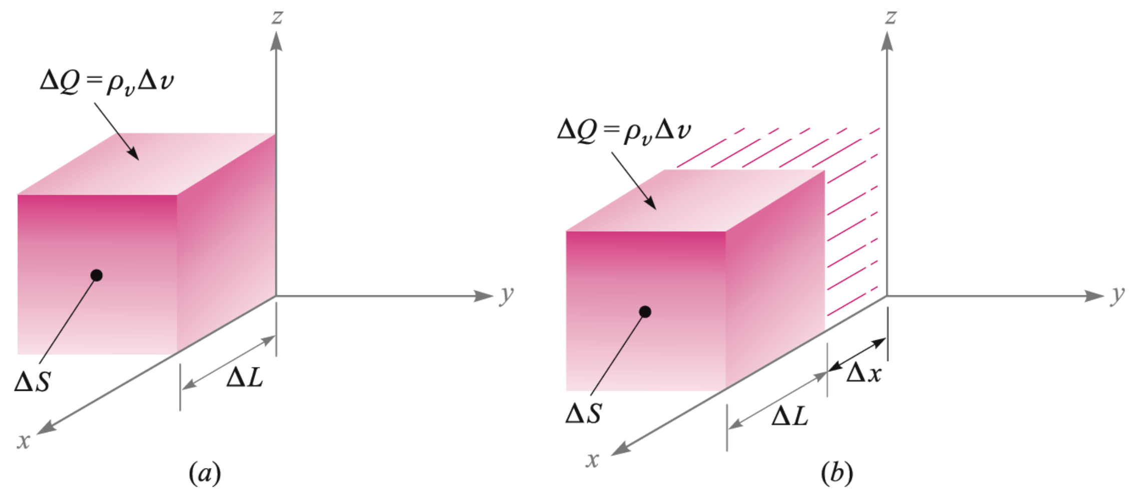
- Relation of Current to Charge Velocity

So we now have $\Delta I = \rho_v \, \Delta S \, v_x$

The current density is then: $$J_x = \frac{\Delta I}{\Delta S} = \rho_v v_x$$

So that in general:

$$\boxed{\mathbf{J} = \rho_v \mathbf{v}}$$



Q: How to define the direction of **v** herein ?

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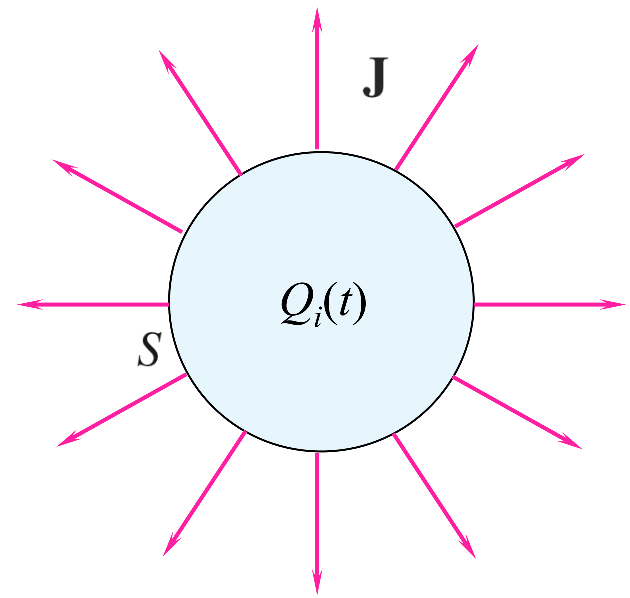
- Continuity of Current 电流的连续性

Suppose that charge Q_i is escaping from a volume through closed surface S, to form current density **J**. Then the total current is:

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_i}{dt} = -\frac{d}{dt}\int_{\text{vol}} \rho_v \, dv$$

where the minus sign is needed to produce *positive* outward flux, while the interior charge is *decreasing* with time.



We now apply the divergence theorem:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \int_{\text{vol}} (\nabla \cdot \mathbf{J})\, dv$$

so that $\displaystyle\int_{\text{vol}} (\nabla \cdot \mathbf{J})\, dv = -\frac{d}{dt}\int_{\text{vol}} \rho_v \, dv$

or $\displaystyle\int_{\text{vol}} (\nabla \cdot \mathbf{J})\, dv = \int_{\text{vol}} -\frac{\partial \rho_v}{\partial t}\, dv$

The integrands of the last expression must be equal, leading to the *Equation of Continuity* 连续性方程

$$\boxed{(\nabla \cdot \mathbf{J}) = -\frac{\partial \rho_v}{\partial t}}$$

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- Example

Consider a current density that is directed radially outward and decreases exponentially with time,

$$\mathbf{J} = \frac{1}{r} e^{-t} \mathbf{a}_r \ \text{A/m}^2$$

Selecting an instant of time $t = 1$ s, we may calculate the total outward current at $r = 5$ m:

$$I = J_r S = \left(\tfrac{1}{5} e^{-1}\right)(4\pi 5^2) = 23.1 \text{ A}$$

At the same instant, but for a slightly larger radius, $r = 6$ m, we have

$$I = J_r S = \left(\tfrac{1}{6} e^{-1}\right)\left(4\pi 6^2\right) = 27.7 \text{ A}$$

Thus, the total current is larger at $r = 6$ than it is at $r = 5$.

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = \nabla \cdot \left(\frac{1}{r} e^{-t} \mathbf{a}_r\right) = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{1}{r} e^{-t}\right) = \frac{1}{r^2} e^{-t}$$

9: Conductors & Dielectrics

- Example

$$-\frac{\partial \rho_v}{\partial t} = \nabla \cdot \mathbf{J} = \nabla \cdot \left(\frac{1}{r}e^{-t}\mathbf{a}_r\right) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{1}{r}e^{-t}\right) = \frac{1}{r^2}e^{-t}$$

$$\rho_v = -\int \frac{1}{r^2}e^{-t}\,dt + K(r) = \frac{1}{r^2}e^{-t} + K(r)$$

If we assume that $\rho_v \to 0$ as $t \to \infty$, then $K(r) = 0$, and

$$\rho_v = \frac{1}{r^2}e^{-t}\ \text{C/m}^3$$

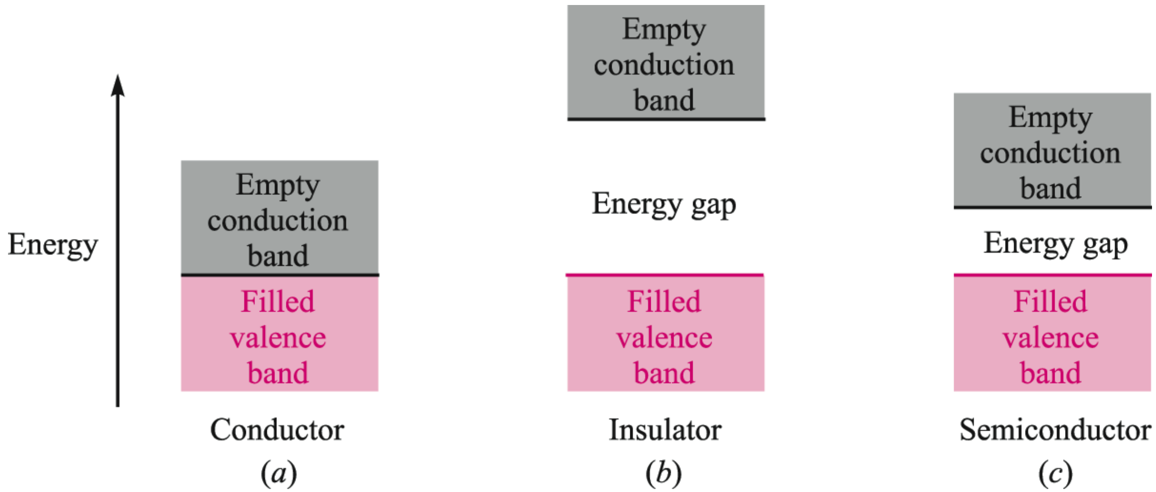
We may now use $\mathbf{J} = \rho_v \mathbf{v}$ to find the velocity,

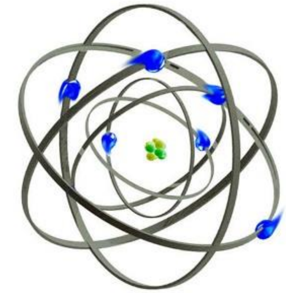
$$v_r = \frac{J_r}{\rho_v} = \frac{\frac{1}{r}e^{-t}}{\frac{1}{r^2}e^{-t}} = r\ \text{m/s}$$

9: Conductors & Dielectrics

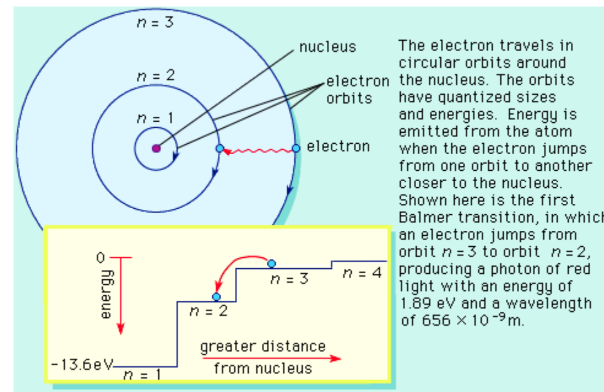
- Metallic Conductors 金属导体

Energy Band Structure in Three Material Types





Rutherford's Atomic Model



Bohr Atomic Model

a) Conductors exhibit no energy gap between valence and conduction bands, so electrons move freely.

b) Insulators show large energy gaps, requiring large amounts of energy to lift electrons into the conduction band. When this occurs, the dielectric breaks down.

c) Semiconductors have a relatively small energy gap, so modest amounts of energy (applied through heat, light, or an electric field) may lift electrons from valence to conduction bands.

9: Conductors & Dielectrics

• Electron Flow in Conductors

Free electrons move under the influence of an electric field. The applied force on an electron of charge Q = -e will be

$$\mathbf{F} = -e\mathbf{E}$$

When forced, the electron accelerates to an equilibrium velocity, known as the *drift velocity* 漂移速度:

$$\mathbf{v}_d = -\mu_e \mathbf{E}$$

where μ_e is the electron *mobility* 漂移率, expressed in units of $m^2/(Vs)$. The drift velocity is used to find the current density through:

$$\mathbf{J} = \rho_v \mathbf{v}_d = -\rho_e \mu_e \mathbf{E} = \sigma \mathbf{E} \qquad \boxed{\mathbf{J} = \sigma \mathbf{E}}$$

from which we identify the *conductivity* 电导率 for the case of electron flow :

$$\boxed{\sigma = -\rho_e \mu_e} \quad \text{S/m}$$

9: Conductors & Dielectrics

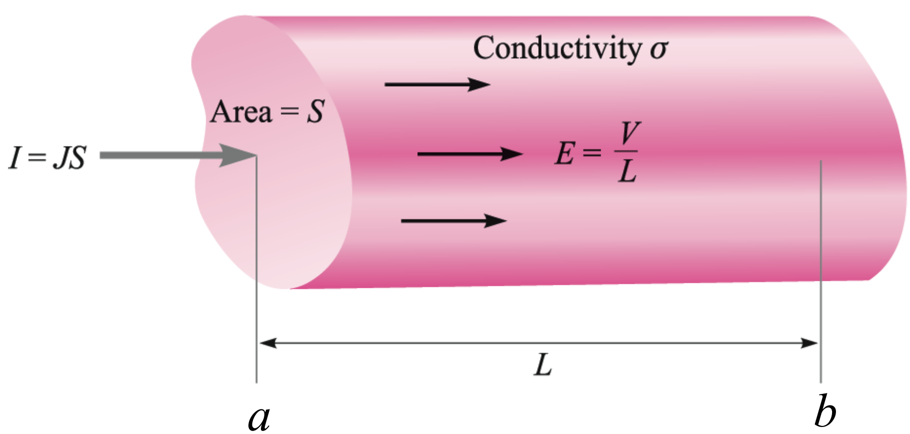
- Conductance & Resistance 电导&电阻

Consider the cylindrical conductor shown here, with voltage V applied across the ends. Current flows down the length, and is assumed to be uniformly distributed over the cross-section, S.

First, we can write the voltage and current in the cylinder in terms of field quantities:

$$V_{ab} = -\int_b^a \mathbf{E} \cdot d\mathbf{L} = -\mathbf{E} \cdot \int_b^a d\mathbf{L} = -\mathbf{E} \cdot \mathbf{L}_{ba} = \mathbf{E} \cdot \mathbf{L}_{ab} \Longrightarrow V = EL$$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} = JS \quad \Longrightarrow \quad J = \frac{I}{S} = \sigma E = \sigma \frac{V}{L} \Longrightarrow \underline{V = \frac{L}{\sigma S} I}$$



Using Ohm's Law:

$$\boxed{V = IR}$$

We find the resistance of the cylinder:

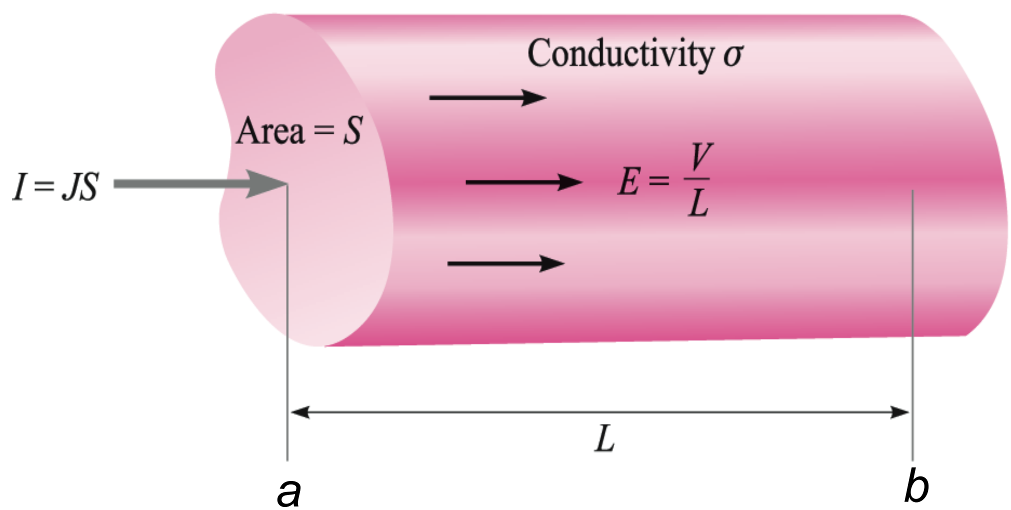
$$\boxed{R = \frac{L}{\sigma S}}$$

Unit: ohm

- Conductance & Resistance 电导&电阻

General Expression for Resistance

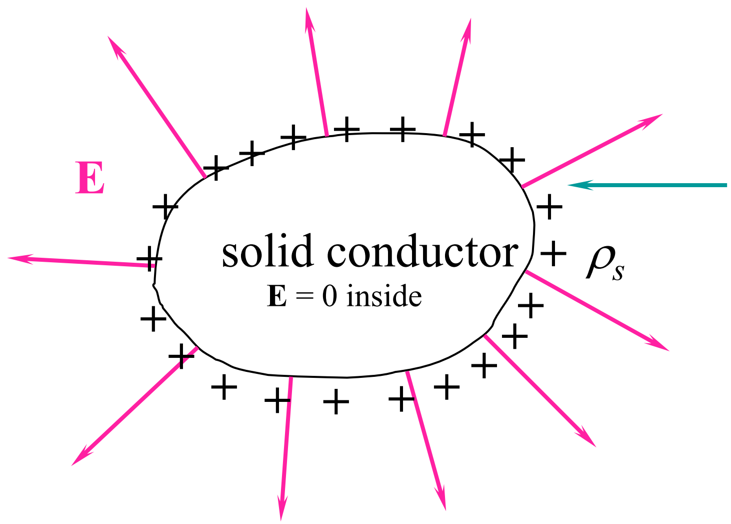
$$\boxed{R = \frac{V_{ab}}{I} = \frac{-\int_b^a \mathbf{E} \cdot d\mathbf{L}}{\int_S \sigma \mathbf{E} \cdot d\mathbf{S}}}$$



9: Conductors & Dielectrics

- Electrostatic Properties of Conductors

Consider a conductor, on which *excess* charge has been placed

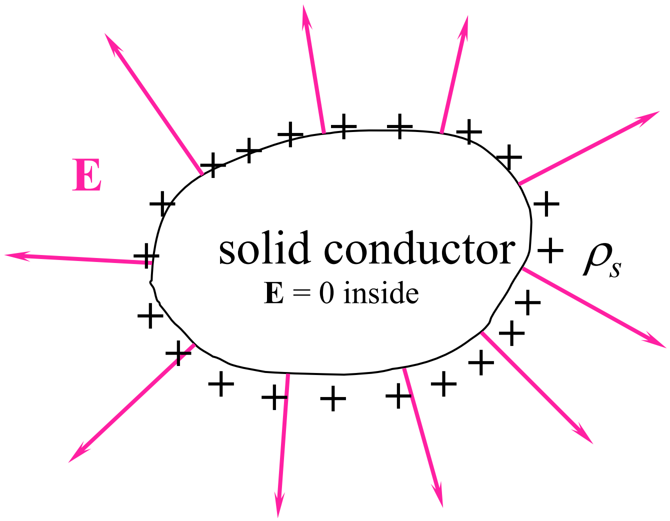


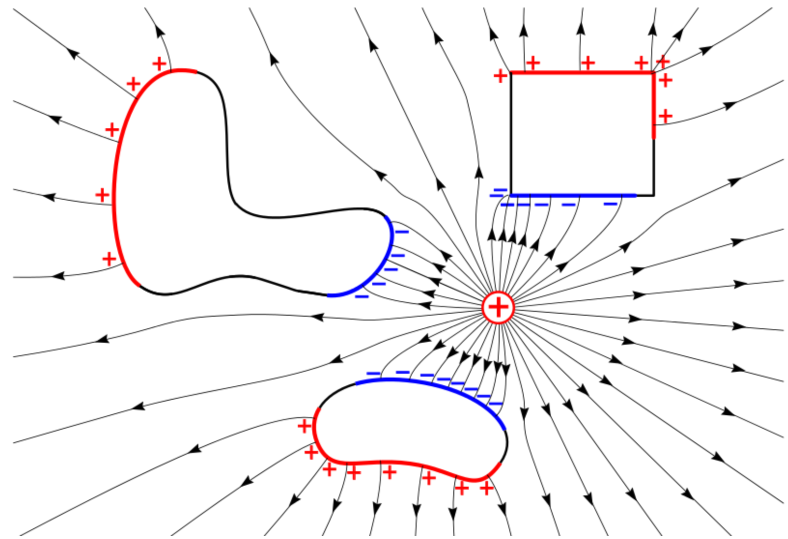
Electric field at the surface points in the *normal* direction

1. Charge can exist only on the surface as a surface charge density, ρ_s -- *not* in the interior.
2. Electric field *cannot* exist in the interior, nor can it possess a tangential component at the surface.
3. It follows from condition 2 that the surface of a conductor is an *equipotential*.

9: Conductors & Dielectrics

- Electrostatic Properties of Conductors





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9: Conductors & Dielectrics

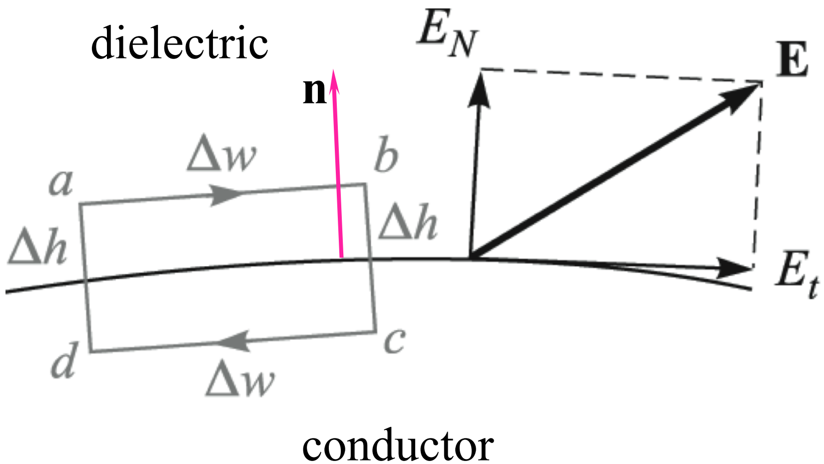
- ## Boundary Condition

Over the rectangular integration path, we use

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0 \quad \text{or} \quad \int_a^b + \int_b^c + \int_c^d + \int_d^a = 0$$

To find:

$$E_t \Delta w - \underbrace{E_{N,\text{at}\,b} \tfrac{1}{2}\Delta h + E_{N,\text{at}\,a} \tfrac{1}{2}\Delta h}_{\text{These become negligible as } \Delta h \text{ approaches zero.}} = 0$$



Therefore

$$\boxed{E_t = 0}$$

More formally: $\boxed{\mathbf{E} \times \mathbf{n}\big|_s = 0}$

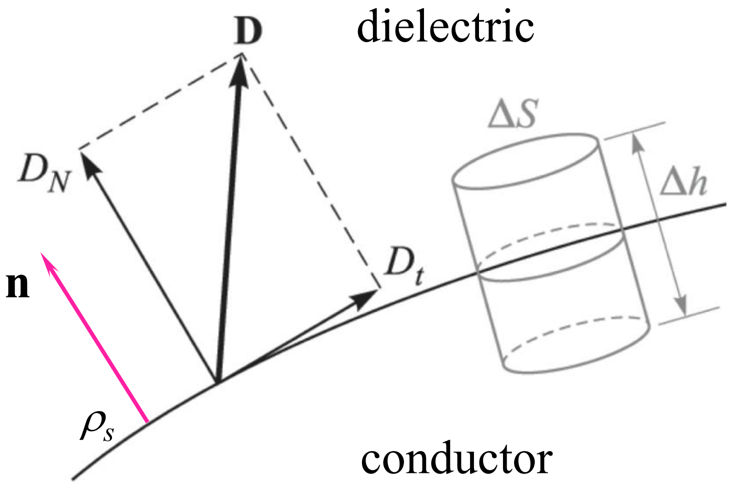
9: Conductors & Dielectrics

- Boundary Condition

Gauss' Law is applied to the cylindrical surface shown below:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{top}} + \cancel{\int_{\text{bottom}}} + \cancel{\int_{\text{sides}}} = Q$$

This reduces to: $D_N \Delta S = Q = \rho_S \Delta S$ as Δh approaches zero



Therefore

$$\boxed{D_N = \rho_S}$$

More formally:

$$\boxed{\mathbf{D} \cdot \mathbf{n}\Big|_s = \rho_s}$$

9: Conductors & Dielectrics

- Summary

1. The static electric field intensity inside a conductor is zero.

2. The static electric field intensity at the surface of a conductor is everywhere directed normal to that surface.

3. The conductor surface is an equipotential surface.

At the surface:

$$\mathbf{E} \times \mathbf{n}\Big|_S = 0$$ Tangential **E** is zero

$$\mathbf{D} \cdot \mathbf{n}\Big|_S = \rho_S$$ Normal **D** is equal to the surface charge density

9: Conductors & Dielectrics

- Example

Given the potential,

$$V = 100(x^2 - y^2)$$

and a point $P(2, -1, 3)$ that is stipulated to lie on a conductor-to-free-space boundary, find V, $\mathbf{E}$, $\mathbf{D}$, and ρ_S at P, and also the equation of the conductor surface.

9: Conductors & Dielectrics

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Solution. The potential at point P is

$$V_P = 100[2^2 - (-1)^2] = 300 \text{ V}$$

Because the conductor is an equipotential surface, the potential at the entire surface must be 300 V. Moreover, if the conductor is a solid object, then the potential everywhere in and on the conductor is 300 V, for $\mathbf{E} = 0$ within the conductor.

The equation representing the locus of all points having a potential of 300 V is

$$300 = 100(x^2 - y^2)$$

or

$$x^2 - y^2 = 3$$

9: Conductors & Dielectrics

- Example

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and a point $P(2, -1, 3)$ that is stipulated to lie on a conductor-to-free-space boundary, find V, $\mathbf{E}$, $\mathbf{D}$, and ρ_S at P, and also the equation of the conductor surface.

Next, we find $\mathbf{E}$ by the gradient operation,

$$\mathbf{E} = -100\nabla(x^2 - y^2) = -200x\mathbf{a}_x + 200y\mathbf{a}_y$$

At point P,

$$\mathbf{E}_p = -400\mathbf{a}_x - 200\mathbf{a}_y \text{ V/m}$$

Because $\mathbf{D} = \epsilon_0\mathbf{E}$, we have

$$\mathbf{D}_P = 8.854 \times 10^{-12}\mathbf{E}_P = -3.54\mathbf{a}_x - 1.771\mathbf{a}_y \text{ nC/m}^2$$

The field is directed downward and to the left at P; it is normal to the equipotential surface. Therefore,

$$D_N = |\mathbf{D}_P| = 3.96 \text{ nC/m}^2$$

Thus, the surface charge density at P is

$$\rho_{S,P} = D_N = 3.96 \text{ nC/m}^2$$

9: Conductors & Dielectrics

- Example

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$$V = 100(x^2 - y^2)$$

and a point $P(2, -1, 3)$ that is stipulated to lie on a conductor-to-free-space boundary, find V, $\mathbf{E}$, $\mathbf{D}$, and ρ_S at P, and also the equation of the conductor surface.

Finally, let us determine the equation of the streamline passing through P.

Solution. We see that

$$\frac{E_y}{E_x} = \frac{200y}{-200x} = -\frac{y}{x} = \frac{dy}{dx}$$

Thus, $$\frac{dy}{y} + \frac{dx}{x} = 0$$

$$\ln y + \ln x = C_1$$

Therefore, $$xy = C_2$$

The line (or surface) through P is obtained when $C_2 = (2)(-1) = -2$

Thus, $$xy = -2$$